

Linear algebra – 11

Non-negative matrices: Perron and Frobenius

Homework

12 сентября 2026 года

Theoretical minimum. Non-negative, irreducible and primitive matrices; the theorems of Perron and Frobenius; the Collatz–Wielandt formula; stochastic and doubly stochastic matrices; the Birkhoff–von Neumann theorem; the adjacency matrix of a graph and its spectrum; the permanent.

1. Prove Frobenius' theorem for an irreducible non-negative matrix, and show that the period h equals the greatest common divisor of the lengths of the cycles of the digraph.
2. (Wielandt 1950) Prove that a primitive $n \times n$ matrix satisfies $A^{(n-1)^2+1} > 0$, and show that the exponent $(n-1)^2+1$ cannot be lowered.
3. (IMC) Let A be a symmetric invertible $n \times n$ matrix ($n \geq 2$) with strictly positive entries and let z_n be the number of zero entries of A^{-1} .
 - a) Show that $z_n \leq n^2 - 2n$.
 - b) Count the zero entries of the inverse of the matrix A with $a_{ij} = 1$ when $\min(i, j)$ is odd and $a_{ij} = 2$ when $\min(i, j)$ is even, and check that the bound of a) is attained.
4. (Высшая лига 2020) Let A be the complex $n \times n$ matrix whose first $n-1$ rows are $(0, 1, 0, \dots, 0)$, $(0, 0, 1, \dots, 0)$, $\dots$ and whose last row is $(1/n, 1/n, \dots, 1/n)$. Is it true that $\lim_{k \rightarrow \infty} A^k$ exists and is a projection onto a one-dimensional subspace? If the limit exists, find its kernel and its image.

5. (Birkhoff–von Neumann) Prove that every doubly stochastic matrix is a convex combination of permutation matrices, and identify the vertices of the Birkhoff polytope. (*Hint: Hall's marriage theorem.*)
6. Prove that an irreducible aperiodic Markov chain has a unique stationary distribution $\pi > 0$ and that the distribution at time k converges to π geometrically, with rate given by the second largest eigenvalue modulus.
7. (PageRank) Let S be a column-stochastic matrix, $\alpha \in (0, 1)$ and $G = \alpha S + (1 - \alpha) \frac{1}{n} J$, where J is the all-ones matrix. Prove that G has a unique positive stationary vector and that the power iteration converges with rate α .
8. (SEEMOUS 2009) Given real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$, define $A = (a_{ij})$ by $a_{ij} = a_i - b_j$ and $B = (b_{ij})$ by $b_{ij} = 1$ if $a_{ij} \geq 0$ and $b_{ij} = 0$ otherwise. Let $C = (c_{ij})$ be a 0/1 matrix with the same row sums and the same column sums as B . Show that
 - a) $\sum_{i,j} a_{ij}(b_{ij} - c_{ij}) = 0$ and $B = C$;
 - b) B is invertible if and only if there are permutations σ, τ with $b_{\tau(1)} \leq a_{\sigma(1)} < b_{\tau(2)} \leq a_{\sigma(2)} < \dots < b_{\tau(n)} \leq a_{\sigma(n)}$.
9. Let G be a graph with adjacency matrix A , average degree $\bar{d}$ and maximum degree Δ . Prove that $\bar{d} \leq \lambda_{\max}(A) \leq \Delta$, that G is bipartite if and only if its spectrum is symmetric about zero, and that the number of closed walks of length k equals $\text{tr}(A^k)$.
10. Prove that $\text{per}(A) \geq 0$ for $A \geq 0$ but that the permanent is not multiplicative, state the van der Waerden bound $\text{per}(A) \geq n!/n^n$ for doubly stochastic A , and check it for $A = \frac{1}{n}J$ and for permutation matrices.
11. (IMO shortlist 2003) Let a_{ij} , $i, j = 1, 2, 3$, be real numbers such that a_{ij} is positive for $i = j$ and negative for $i \neq j$. Prove that there exist positive real numbers c_1, c_2, c_3 such that the numbers

$$a_{11}c_1 + a_{12}c_2 + a_{13}c_3, \quad a_{21}c_1 + a_{22}c_2 + a_{23}c_3, \quad a_{31}c_1 + a_{32}c_2 + a_{33}c_3$$

are all negative, all positive, or all zero.

12. (Putnam 1995) Let $x_1, \dots, x_n$ be differentiable real-valued functions of a single variable t satisfying $\frac{dx_i}{dt} = \sum_{j=1}^n a_{ij}x_j$ for constants $a_{ij} > 0$. Suppose that $x_i(t) \rightarrow 0$ as $t \rightarrow \infty$ for every i . Are the functions $x_1, \dots, x_n$ necessarily linearly dependent?